

Sai Praneet Reddy Chinthala

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EDUCATION

George Mason University

M.S. Data Analytics Engineering — GPA: 3.97/4.0

Fairfax, Virginia

Aug 2024 – May 2026

Mahindra University

B.Tech, Electronics and Computer Engineering

Hyderabad, India

Sept 2020 – May 2024

SKILLS & TECHNOLOGIES

ML & GenAI: RAG pipelines, LLMs, prompt engineering, NLP, SBERT, FLAN-T5, LangChain, LangGraph, FAISS, ChromaDB, BM25, Cohere Rerank, Langfuse, PyTorch, TensorFlow, scikit-learn, XGBoost

Data Engineering: Python, SQL, R, feature engineering, data cleansing, pandas, NumPy, Apache Spark, PySpark, ETL/ELT, MySQL, MongoDB

Cloud & DevOps: AWS (S3, EC2, Lambda, DynamoDB), AWS SAM, Azure, Docker, CI/CD (GitHub Actions), Git, Databricks SQL, Power BI

WORK EXPERIENCE

Data Engineer (Volunteer)

Saayam For All — 501(c)(3) non-profit

Aug 2026 – Present

Remote

- Contribute to the data-analytics workstream: **ETL and data-quality tooling** supporting BI reporting.
- Built a **synthetic data generator** and validation suite over ten relationally consistent tables with foreign-key and geo-coordinate checks, enabling dashboard work without exposing real user records.
- Reconciled the schema across three sources and flagged **three live discrepancies** between the team's wiki and production data.

Graduate Teaching Assistant (AIT 614 — Big Data Essentials)

George Mason University

Jan 2026 – May 2026

Fairfax, Virginia

- Led **10 hands-on labs** across two sections covering **Databricks, Spark/PySpark, MongoDB, Hadoop, AWS** and **LangChain-based RAG**.
- Graded **100+ Python and PySpark submissions** from **35 students** on ETL, model training and feature engineering.
- Corrected course labs used as the student reference solutions and mentored teams through term-long big-data projects.

Generative AI Engineer Intern

Mahindra & Mahindra Ltd

Feb 2024 – May 2024

Remote

- Consolidated **20+ machine-breakdown root-cause Excel logs** from three plants into one standardised dataset using **Python (pandas/openpyxl)**.
- Adapted an internal **RAG chatbot** (OpenAI embeddings + vector search) to the cleaned corpus so engineers could query breakdown history in natural language.
- Tuned data preparation and prompt formatting to improve answer relevance for maintenance queries.

PROJECTS

DAPSE: Arctic Policy Intelligence Engine

Master's Capstone (team of 6) — Product Owner

Jan 2026 – May 2026

Code private

- Co-developed a **RAG engine** with **BM25 + FAISS hybrid retrieval**, Reciprocal Rank Fusion and Cohere Rerank over **955+ documents**; **nDCG@5 0.832, Precision@5 0.954**.
- Directed delivery of a **verify-then-escalate** agent loop and an **8-dimension risk-scoring engine**, with Langfuse observability and Docker deployment.

Senior Care RAG Agent | [GitHub](#)

Aug 2026

- Built an **agentic RAG assistant** with a **query → retrieve → grade → generate → verify** pipeline; every answer grounded and cited.
- 20-question eval set: **Hit@4 0.950, MRR 0.783, faithfulness 0.950**. **LangGraph**, FAISS/ChromaDB; served via FastAPI, Streamlit and an MCP server, with Docker and CI.

Bridges-NBI-Analysis | [GitHub](#)

Aug 2025 – Dec 2025

- Built a **PySpark ETL + ML pipeline** over **950K+ National Bridge Inventory records**; engineered **26 features from 15 source items** and handled a 9:1 class imbalance.
- Tuned **XGBoost from 0.75 to 0.93 ROC-AUC** and identified **truck traffic as the strongest predictor**; persisted to MongoDB with automated Databricks SQL reporting.

Healthcare Financial Analysis | [GitHub](#)

Aug 2024 – Dec 2024

- Built a multi-tool **ETL pipeline** (Python, MySQL, R) over **4,233 CMS Hospital Cost Report records**; quantified the charity-care versus net-income tradeoff across U.S. hospitals.

RESEARCH

Fair Bilevel Optimization for Collaborative Classification

George Mason University, Dept. of Information Sciences & Technology

Jan 2026 – Present

Fairfax, Virginia

- Developing a **bilevel optimization framework** with **Augmented Lagrangian methods** to enforce fairness constraints in **federated learning** across institutions.